Esempi e esercizi guidati

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Esempio 2 – Dinamica longitudinale di un autoveicolo

The Simulink project has to be split into 3 files:

- **Harness.slx**, containing reference models for the controller and plant and test stimuli generation
- **Controller.slx**, containing the controller (to be developed)
- **Plant.slx**, containing the car longitudinal physical model.

The longitudinal physical model and the test stimuli generators shall demonstrate the effectiveness of the implemented controller with respect to the expected functionalities.

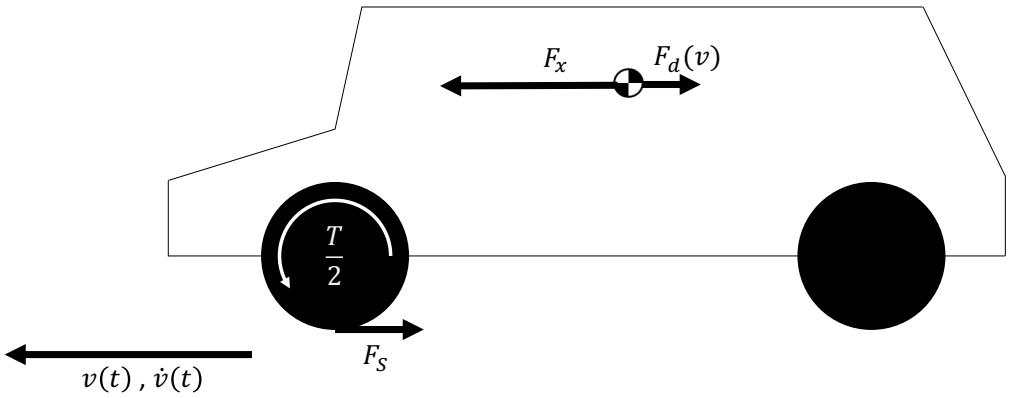
External interfaces of the plant

Name	Direction	Type
Requested_Torque_Nm	Input	CAN
Vehicle_Speed_km_h	Output	CAN
Automatic_Transmission_Selector	Input (from the driver to the controller)	CAN {P, R, N, D, P}
Selected mode/errors	Output (to the driver)	CAN

Equations of the plant

The plant considered in this model is the so-called *Vehicle Longitudinal Dynamics*.

Considering:



- $\dot{v}(t)$ the vehicle acceleration, expressed in $[\mathrm{m/s^2}]$
- $v(t)$ the vehicle longitudinal speed, expressed in $[\mathrm{m/s}]$
- m the vehicle mass, expressed in $[\mathrm{kg}]$
- $F_x(t)$ the longitudinal force applied to the vehicle center of gravity, expressed in [N]
- $F_s(t)$ the longitudinal force applied to the wheel on the terrain, expressed in [N]
- $F_d(v)$ the longitudinal force applied to the vehicle center of gravity due to the frictions with air and terrain, expressed in [N]
- I the moment of inertia of each one of the wheels, expressed in $[\mathrm{kg} \cdot \mathrm{m^2}]$
- r the radius of the wheel, expressed in $[\mathrm{m}]$
- $\omega(t)$ is the angular speed of the wheel, expressed in $[\mathrm{rad/s}]$
- $\omega_e(t)$ is the angular speed of the engine/electrical motor, expressed in $[\mathrm{rad/s}]$
- $\omega_{e_{RPM}}(t) = \frac{\omega_e(t) \cdot 60}{2\pi}$ is the angular speed of the engine/ electrical motor, expressed in $[\mathrm{revolutions\ per\ minutes}]$
- $\dot{\omega}(t)$ is the angular acceleration of the wheel, expressed in $[\mathrm{rad/s^2}]$
- S is the frontal surface of the car, expressed in $[\mathrm{m^2}]$
- $c_x = 0.3$ is the automobile drag coefficient
- ρ is the average density of air at sea level in standard conditions → $\rho = 1.25\ \mathrm{kg/m^3}$
- i_g is the gearbox reduction ratio
- i_f is the final drive reduction ratio
- $i_t = i_g \cdot i_f$ is the total power train reduction ratio.

An extremely simplified model can be obtained as follow:

$$\dot{v}(t) = \frac{F_x(t) - F_d(v)}{m} \qquad (1)$$

where $\dot{v}(t)$ is the vehicle acceleration, m is its mass, $F_x(t)$ is the longitudinal force applied to its center of gravity by the effects of the torque applied on the wheels, and $F_d(v)$ is the sum of the friction forces on the vehicle due to wheel-terrain and vehicle-air interactions.

Considering that the torque is equally split between the two wheels (valid only on straight tracks)

$$T(t) - 2 \cdot F_s(t) \cdot r = 2I \cdot \dot{\omega}(t) \quad (2)$$

the absence of slipping:

$$\begin{cases} 2 \cdot F_s(t) \cdot r = F_x(t) \\ v(t) = 2\,\pi \cdot r \cdot \omega(t) \end{cases}$$

and considering the moment of inertia of the wheels $I = 0$, we can define the following equation, given that $T(t) = 2 \cdot F_s(t) \cdot r \rightarrow F_s(t) = \frac{T(t)}{2 \cdot r}$.

The drag force that limits the maximum speed of the vehicle is equal to:

$$F_d\big(v(t)\big) = X_{air} \cdot \big(v(t)\big)^2 + X_{tyres} \cdot v(t) \quad (3)$$

where:

$$X_{air} = \frac{1}{2} \cdot S \cdot c_x \cdot \rho \quad (4)$$

and, as usually modeled:

$$X_{tyres}\big|_{X_{tyres}\left(50\frac{\mathrm{km}}{\mathrm{h}}\right) = X_{air}\left(50\frac{\mathrm{km}}{\mathrm{h}}\right)} \rightarrow X_{tyres} = \frac{X_{air} \cdot 50}{3.6} \quad (5)$$

By substituting the (2) equation in (1), and by integrating both sides, we obtain:

$$v(t) = \frac{1}{m}\int_0^t F_x(t) - F_d\big(v(t)\big)\,dt = \frac{1}{m}\int_0^t 2 \cdot F_s(t) - F_d\big(v(t)\big)\,dt =$$

$$= \frac{1}{m}\int_0^t \frac{T(t)}{r} - F_d\big(v(t)\big)\,dt \quad (6)$$

and, by substituting (3) in (6):

$$v(t) = \frac{1}{m}\int_0^t \frac{T(t)}{r} - X_{air} \cdot \big(v(t)\big)^2 - X_{tyres} \cdot v(t)\,dt \quad (7)$$

Remember that the integrator block of Simulink requires an initial condition corresponding to the vehicle's longitudinal speed at the beginning of the simulation, $v(0)$. A possible configuration of the integration block is shown in Figure 2.

During the model development, put all the needed gain to obtain as an output of the physical model a speed expressed in km/h.

To simulate the slope θ of the terrain, it is possible to add the gravity force $F_g(\theta)$ as follows:

$$v(t) = \frac{1}{m}\int_0^t \frac{T(t)}{r} - X_{air} \cdot \left(v(t)\right)^2 - X_{tyres} \cdot |v(t)| \cdot dt + m \cdot g \cdot \sin(\theta) \quad (8)$$

With $g = 9.81\ \mathrm{m/s^2}$ the gravity acceleration on Earth.

Reasonable values for an electric compact car can be:

- $m = 1600\ \mathrm{kg}$
- $r = 0.3\ \mathrm{m}$
- The torque T (at the wheel) can vary in the range $[-60; 960]\ \mathrm{Nm}$
- $S = 3.5\ m^2$
- $c_x = 0.3$

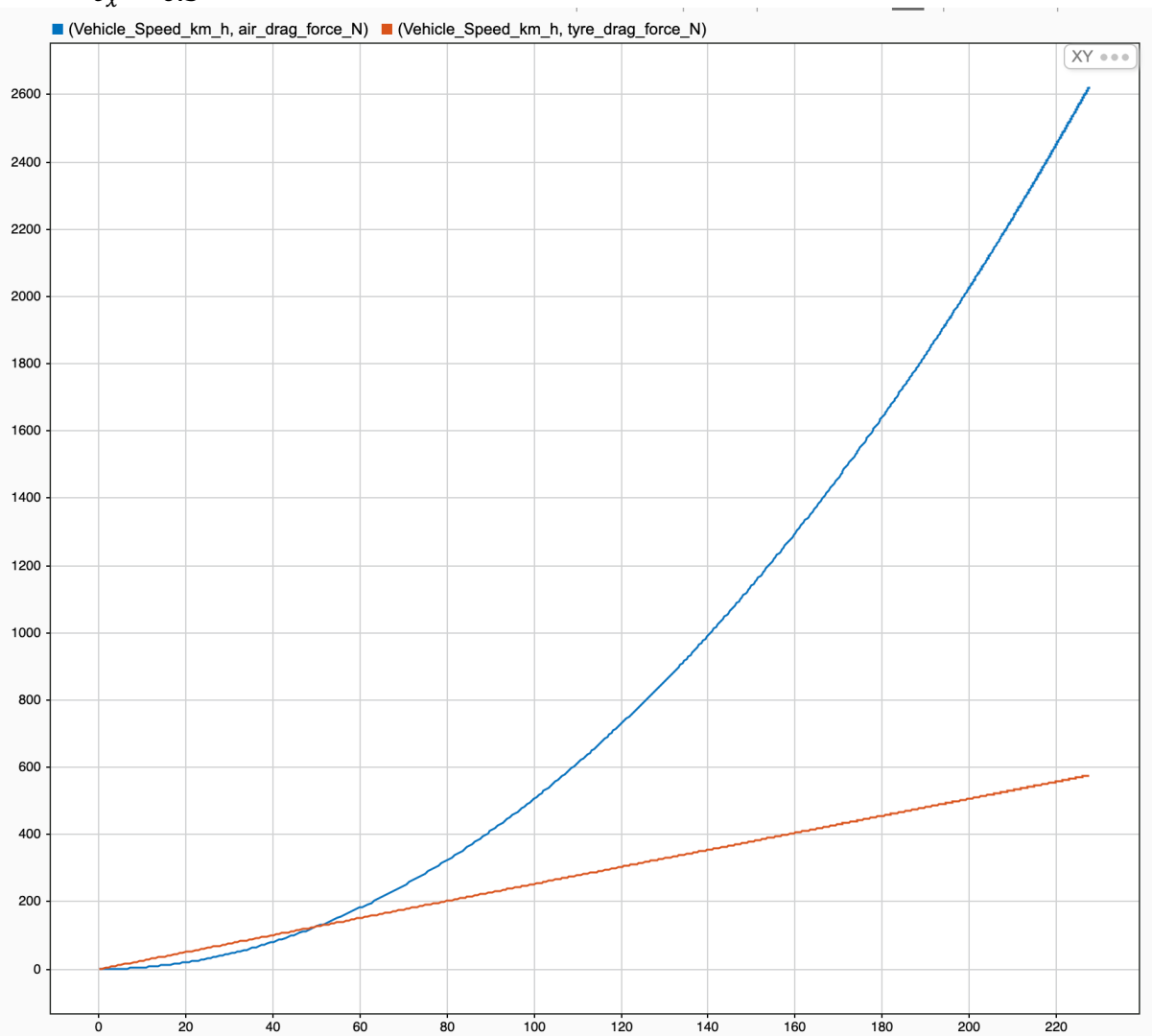


Figure 1 Graph showing drag forces of tires (in orange) and air (in blue) at various speeds. It is possible to observe that, as imposed in equation (5), $X_{tyre} = X_{air}$ at 50 km/h. Below this speed, the tire drag is dominant, after that, the air drag is dominant. Moreover, it is possible to see the top speed of the car (around 230 km/h) when $F_x = \frac{T}{r} = X_{tyre} + X_{air}$, with $F_x = 3200\ N$.

With those values, the top speed on level ground reachable by the car is about 230 km/h, where the drag forces equal the traction force (3200 N).

Considering the reverse direction, the maximum speed reachable with a limitation of -60 Nm is about 45 km/h.

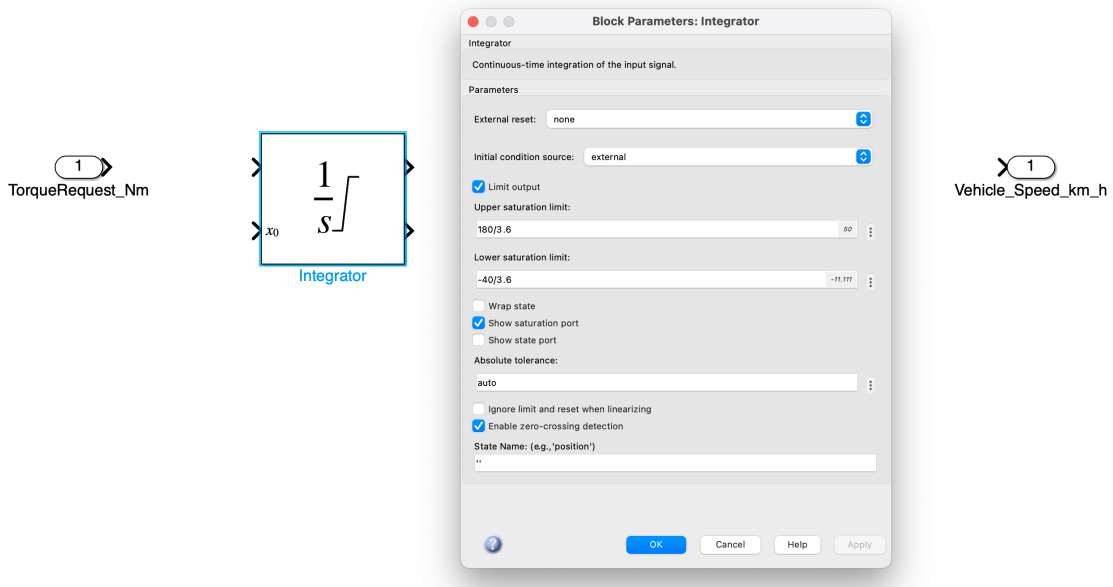


Figure 2 Settings window for the Integrator block of Simulink

Use these values (with a certain tolerance, for example, 10 %) to saturate the integrator block.

To make the model more realistic, it is possible to compute the torque request at the engine/motor. A typical ratio value for transmission of an electric car with a single gear can be around $i_t = 12$[1].

All the initialization parameters of the model are automatically loaded model by a callback of the function **init_fn** as shown in Figure 3.

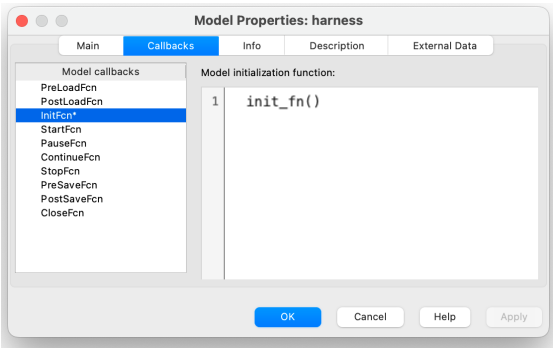


Figure 3 init_fn callback configuration in the harness model properties.

[1] Usually, the first gear of a car has a transmission ratio about $i_g = 3$, hence the engine makes 3 complete rotations very single rotation of the pinion gear of the differential. In the differential (final drive), the pinion gear makes about $i_f = 4$ revolutions for every revolution of the axle shafts. From here, the typical ratio of $i_t = 12$ between the torque at the wheel and the torque at the engine/electrical motor.

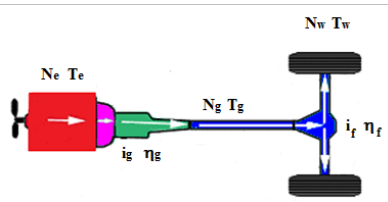


Figure from http://www.thecartech.com/subjects/auto_eng/Gearbox_ratios3.htm